

DS from (N): dominance support and the exact s -valuation as a corollary of the $\mathcal{N}$ -transport

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Summary. You predicted that the DS valuation $\text{val}_s c_{\lambda\mu} = n(\mu)$ would fall out of (N). It does. Support, lead, exact valuation and a closed formula for the lowest coefficient

$$d_{\lambda\mu}(t) = \sum_{\rho} t^{n(\rho) - n(\lambda')} \tilde{K}_{\rho\mu'}(t) K_{\rho'\lambda}$$

follow from (N) in about 15 lines. The minimal s -order is attained by a single term, so there is no cancellation. (N) does *not* see polynomiality of $c_{\lambda\mu}$ or regularity at $t = 0$, so the elementary Day 214 proof is still needed for the integrality half. Computer checks are complete for $n \leq 4$; $n = 5$ is in progress.

0. Conventions

- $E_k F = \sum_{|A|=k} \prod_{i \in A, j \notin A} \frac{x_i - tx_j}{x_i - x_j} X_A F(X_{A^c}, sX_A)$; $e_{\lambda}^* := E_{\lambda_1} \cdots E_{\lambda_\ell}(1) = \sum_{\mu} c_{\lambda\mu} e_{\mu}$; $n(\lambda) = \sum (i-1)\lambda_i$.
- $P_{\nu} := P_{\nu}(x; q = s, t_{\text{Mac}} = \tau)$ with $\tau := 1/t$; $T_{\nu} := t^{n(\nu)} s^{n(\nu')}$; $\mathcal{N}P_{\nu} = T_{\nu}P_{\nu}$.
- (N) (Day 216c, proved modulo the Cherednik facts C1–C3): $E_k = t^{-\binom{k}{2}} \mathcal{N}e_k \mathcal{N}^{-1}$. Since $\mathcal{N}(1) = 1$,

$$(0.1) \quad e_{\lambda}^* = t^{-\sum \binom{\lambda_i}{2}} \mathcal{N}(e_{\lambda}) = t^{-n(\lambda')} \mathcal{N}(e_{\lambda}).$$

That is, $\Psi = \mathcal{N}^{-1}$ with $\Psi(e_{\lambda}^*) = t^{-n(\lambda')} e_{\lambda}$. The input to $\mathcal{N}$ is e_{λ} , not e_{λ}^* .

- $\mathcal{R} := \mathbb{Q}(t)[s]_{(s)}$, rational functions in s over $\mathbb{Q}(t)$ regular at $s = 0$; val_s is taken in $\mathbb{Q}(t)(s)$.

1. Two triangular change-of-basis matrices

Lemma 1 (P in e). Write $P_{\nu} = \sum_{\mu} B_{\nu\mu} e_{\mu}$. Then $B_{\nu\mu} \neq 0 \Rightarrow \mu \supseteq \nu'$; $B_{\nu\nu'} = 1$; $B_{\nu\mu} \in \mathcal{R}$; and $B_{\nu\mu}|_{s=0}$ is the e -expansion of Hall–Littlewood $P_{\nu}(x; \tau)$.

Proof. (1) $P_{\nu} = m_{\nu} + \sum_{\kappa \triangleleft \nu} u_{\nu\kappa} m_{\kappa}$ (Macdonald VI (4.7)). (2) By Day 214 Lemma 1.1, $\text{span}\{m_{\kappa} : \kappa \trianglelefteq \nu\} = \text{span}\{e_{\mu} : \mu \supseteq \nu'\}$ with unitriangular integer transition ($m_{\nu} = e_{\nu'} + \text{higher}$). This gives the support and the 1. (3) Regularity at $s = 0$: by the combinatorial formula (VI (7.13')), $P_{\nu} = \sum_T \psi_T x^T$ with ψ_T a product of ratios $b_{\lambda}(\square)/b_{\mu}(\square)$, $b_{\lambda}(\square; q, t) = (1 - q^a t^{l+1})/(1 - q^{a+1} t^l)$. At $q = s$ every factor $1 - s^{a+1} \tau^l$ and $1 - s^a \tau^{l+1}$ is a unit or a nonzero element of $\mathcal{R}$ (for $a = 0$, $1 - \tau^{l+1} \neq 0$). So $u_{\nu\kappa} \in \mathcal{R}$. (4) $P_{\nu}(x; 0, \tau)$ is HL $P_{\nu}(x; \tau)$ (VI (4.14)(iii)). $\square$

Lemma 2 (e in P). Write $e_{\lambda} = \sum_{\nu} A_{\lambda\nu} P_{\nu}$. Then $A_{\lambda\nu} \neq 0 \Rightarrow \nu \leq \lambda'$; $A_{\lambda\lambda'} = 1$; $A_{\lambda\nu} \in \mathcal{R}$; $A_{\lambda\nu}|_{s=0} = a_{\lambda\nu}^{HL}(\tau)$, the coefficient of HL $P_{\nu}(x; \tau)$ in e_{λ} .

Proof. Ordering rows by ν and columns by μ' , B is unitriangular over $\mathcal{R}$, so $A = B^{-1}$ is too, and $A(0) = B(0)^{-1}$. $\square$

Lemma 3 (HL coefficient). $a_{\lambda\nu}^{HL}(\tau) = \sum_{\rho} K_{\rho\nu}(\tau) K_{\rho'\lambda}$ (Kostka–Foulkes times Kostka). Consequently (a) $a_{\lambda\nu}^{HL}(0) = K_{\nu'\lambda}$, which is > 0 iff $\nu \leq \lambda'$; (b) $\deg_{\tau} a_{\lambda\nu}^{HL} \leq n(\nu) - n(\lambda')$, and the $\tau^{n(\nu) - n(\lambda')}$ coefficient is the leading coefficient of $K_{\lambda'\nu}(\tau)$, equal to 1 when $\nu \leq \lambda'$.

Proof. (1) $Q'_\nu(x; \tau) := Q_\nu[X/(1-\tau); \tau]$ is Hall-dual to $P_\nu(x; \tau)$ by the HL Cauchy identity (Macdonald III (4.4)), and $Q'_\nu = \sum_\rho K_{\rho\nu}(\tau) s_\rho$ (III §7). (2) $a_{\lambda\nu}^{HL} = \langle e_\lambda, Q'_\nu \rangle = \sum_\rho K_{\rho\nu}(\tau) \langle e_\lambda, s_\rho \rangle$ and $\langle e_\lambda, s_\rho \rangle = K_{\rho'\lambda}$. (3) (a): $K_{\rho\nu}(0) = \delta_{\rho\nu}$, and $K_{\nu'\lambda} > 0$ iff $\nu' \supseteq \lambda$. (4) (b): $K_{\rho\nu}(\tau) \neq 0$ only if $\rho \supseteq \nu$; it is monic of degree $n(\nu) - n(\rho)$ (charge attains its maximum at exactly one tableau; Macdonald III §6, locator not re-verified). $K_{\rho'\lambda} \neq 0$ forces $\rho \supseteq \lambda'$, so $n(\rho) \geq n(\lambda')$ with equality only at $\rho = \lambda'$; that term has coefficient $1 \cdot K_{\lambda\lambda} = 1$. $\square$

2. The theorem

Theorem (DS from N). Assume (N). For all λ, μ :

$$(2.1) \quad c_{\lambda\mu} = t^{-n(\lambda')} \sum_{\nu: \lambda \leq \nu' \leq \mu} A_{\lambda\nu} s^{n(\nu')} t^{n(\nu)} B_{\nu\mu}.$$

Consequently:

(S) $c_{\lambda\mu} \neq 0$ only if $\mu \supseteq \lambda$.

(L) $c_{\lambda\lambda} = s^{n(\lambda)}$.

(Val) For $\mu \supseteq \lambda$, $c_{\lambda\mu} \in s^{n(\mu)} \mathcal{R}$, and

$$d_{\lambda\mu}(t) := [s^{n(\mu)}] c_{\lambda\mu} = t^{n(\mu') - n(\lambda')} a_{\lambda\mu'}^{HL}(1/t) = \sum_\rho t^{n(\rho) - n(\lambda')} \tilde{K}_{\rho\mu'}(t) K_{\rho'\lambda},$$

where $\tilde{K}_{\rho\nu}(t) := t^{n(\nu) - n(\rho)} K_{\rho\nu}(1/t)$ is cocharge Kostka–Foulkes.

(D) $d_{\lambda\mu} \in \mathbb{Z}[t]$ with $d_{\lambda\mu}(0) = 1$. Hence $\text{val}_s c_{\lambda\mu} = n(\mu)$ *exactly*, and the support of e_λ^* is *exactly* the up-set $\{\mu \supseteq \lambda\}$. With Lascoux–Schützenberger positivity, $d_{\lambda\mu} \in \mathbb{N}[t]$ and $d_{\lambda\mu}(1) = \sum_\rho K_{\rho\mu'} K_{\rho'\lambda} = \langle h_{\mu'}, e_\lambda \rangle = M_{\lambda\mu'}$ (number of 0-1 matrices).

(V) $c_{\lambda\mu}|_{s=1} = \delta_{\lambda\mu}$, given the $s = 1$ facts quoted below.

Proof.

1. **(2.1).** By (0.1) and Lemmas 1–2, $e_\lambda^* = t^{-n(\lambda')} \sum_\nu A_{\lambda\nu} T_\nu P_\nu = t^{-n(\lambda')} \sum_{\nu, \mu} A_{\lambda\nu} T_\nu B_{\nu\mu} e_\mu$; a term is nonzero only if $\nu \leq \lambda'$ and $\mu \supseteq \nu'$, i.e. $\lambda \leq \nu' \leq \mu$.
2. **(S)** is immediate from $\lambda \leq \nu' \leq \mu$.
3. **(L).** For $\mu = \lambda$ only $\nu' = \lambda$ survives: $t^{-n(\lambda')} \cdot 1 \cdot s^{n(\lambda)} t^{n(\lambda')} \cdot 1 = s^{n(\lambda)}$.
4. **(Val).** n is strictly order-reversing on dominance ($n(\rho) = \sum_i (|\rho| - \rho_1 - \dots - \rho_i)$), so $\nu' \leq \mu$ gives $n(\nu') \geq n(\mu)$, with equality only if $\nu' = \mu$. Since $A, B \in \mathcal{R}$, every term of (2.1) lies in $s^{n(\mu)} \mathcal{R}$ and every term with $\nu \neq \mu'$ lies in $s^{n(\mu)+1} \mathcal{R}$. Hence

$$c_{\lambda\mu} \equiv t^{-n(\lambda')} A_{\lambda\mu'}(0) s^{n(\mu)} t^{n(\mu')} B_{\mu'\mu}(0) = s^{n(\mu)} t^{n(\mu') - n(\lambda')} a_{\lambda\mu'}^{HL}(\tau) \pmod{s^{n(\mu)+1}}.$$

Lemma 3 and $t^{n(\mu') - n(\lambda')} K_{\rho\mu'}(1/t) = t^{n(\rho) - n(\lambda')} \tilde{K}_{\rho\mu'}(t)$ give the Kostka sum. **No cancellation is possible:** the minimal s -order is reached by exactly one ν , so the cancellation worry does not arise at the diagonal.

5. **(D).** Each $t^{n(\rho) - n(\lambda')} \tilde{K}_{\rho\mu'}(t)$ is a polynomial since $n(\rho) \geq n(\lambda')$ whenever $K_{\rho'\lambda} \neq 0$. Only $\rho = \lambda'$ has nonzero constant term, namely $\tilde{K}_{\lambda'\mu'}(0) = \text{lead coefficient of } K_{\lambda'\mu'} = 1$ (Lemma 3(b)), times $K_{\lambda\lambda} = 1$; so $d(0) = 1$. (Nonvanishing alone already follows from Lemma 3(a) at $\tau = 0$: $a_{\lambda\mu'}^{HL}(0) = K_{\mu\lambda} > 0$.) $d(1) = \sum_\rho K_{\rho\mu'} K_{\rho'\lambda} = \langle h_{\mu'}, e_\lambda \rangle = M_{\lambda\mu'}$ (Macdonald I (6.6)).
6. **(V).** $P_\nu(x; q, \tau)$ is regular at $q = 1$ with $P_\nu(x; 1, \tau) = e_{\nu'}$ (VI (4.14)(vi); regularity via integrality of J_ν , VI §8, locator not re-verified). So $B(1)$ is the permutation matrix $\nu \mapsto \nu'$, $A(1)$ its inverse, $T_\nu|_{s=1} = t^{n(\nu)}$, and $c(1) = t^{-n(\lambda')} t^{n(\lambda')} \delta = \delta$. $\square$

Operator form (Op-DS), support and lead. Apply the same argument to $E_k e_\mu = t^{-\binom{k}{2}} \mathcal{N} e_k \mathcal{N}^{-1} e_\mu$: $\mathcal{N}^{-1} e_\mu \in \text{span}\{P_\nu : \nu \leq \mu'\}$; $e_k P_\nu \in \text{span}\{P_\kappa : \kappa \leq \nu + 1^k \leq (\mu \cup k)'\}$ (Macdonald Pieri VI (6.24), or monomial triangularity), with $P_{(\mu \cup k)'}$ -coefficient 1 coming from $\nu = \mu'$ alone. So the support is $\{\rho \supseteq \mu \cup k\}$ and the lead is $t^{-\binom{k}{2}} T_{(\mu \cup k)'}/T_{\mu'} = s^{n(\mu \cup k) - n(\mu)} = s^{\sum_i \min(\mu_i, k)}$, using $n((\mu \cup k)') - n(\mu') = \binom{k}{2}$. Not checked separately by script; the Day 214 Op-DS check covers the statement itself.

3. What (N) does *not* give

- **Polynomiality.** (N) yields only $c_{\lambda\mu} \in \mathbb{Q}(t)(s) \cap s^{n(\mu)}\mathcal{R}$. The cancellation of the Macdonald denominators $1 - s^{a+1}\tau^l$ needed for $c_{\lambda\mu} \in \mathbb{Q}[s, t]$ is invisible from (N). It is a genuine extra fact, supplied by the Gauss-lemma argument of Day 214 (§4/§8.1).
- **Regularity at $t = 0$.** $c_{\lambda\mu}(s, 0) = s^{n(\mu)}(1 + s\mathbb{Q}[s])$ needs regularity at $t = 0$ ($\tau = \infty$), which (N) does not see. Day 214 Thm 2(b) remains the proof.

Our reading is that integrality comes from the operator side, in the spirit of the Kirillov–Noumi (1999) raising operators, not from the transport. So the elementary Day 214 proof is still needed for the integrality half.

Further caveats (kept from the source). (1) *Grade inheritance.* Everything here inherits the grade of (N): proved modulo the Cherednik facts (C1)–(C3) (textbook, locators unverified) and 207b’s $(A_k)/(K_k)$. Day 214’s DS proof uses only 207b and elementary degree counts, so it is the *more elementary* proof; this note is a second proof plus a structural explanation, not a replacement. (2) *Textbook locators not re-verified:* VI (7.13’); VI (4.14)(iii),(vi); III §6–7 (Kostka–Foulkes degree/monic/charge, Hall-duality of Q'); integrality of J for $q = 1$ regularity. The script checks their consequences in the range below, not the statements in general. (3) *Novelty.* The d -formula is the Day 215 “ $d = \text{HL/cocharge } t\text{-count}$ ” statement, now *derived* rather than observed. It is $\langle e_\lambda, \cdot \rangle$ applied to a modified HL function; expect it to be folklore-adjacent once (N) is known.

4. Verification

`scripts/day218b/ds_from_N.py` `N` runs over every $\lambda \vdash n \leq N$ and checks symbolically in s, t : (2.1) against Hikita’s e_λ^* from the operator matrices; support, unitriangularity and s -integrality of A, B ; regularity of B at $s = 1$ and $P(1, \tau) = e_\nu$; exact valuation $n(\mu)$ and $d = t^D a^{HL}(1/t)$; the Kostka–Foulkes formula for d (with $K(t)$ computed independently as the Schur→HL matrix); $d \in \mathbb{N}[t]$, $d(0) = 1$, $d(1) = M_{\lambda\mu'}$; $a^{HL}(\tau = 0) > 0$; off-up-set zeros; (V).

Results (`ds_from_N_5.log`): $n = 1$: 15 OK; $n = 2$: 60 OK; $n = 3$: 150 OK; $n = 4$: 375 OK; 0 BAD throughout. **Checks are complete for $n \leq 4$** (375/375 at $n = 4$ with the full check set). The first $n = 5$ run was truncated; it has been relaunched and is **in progress**. No $n = 5$ result is claimed here. $n = 6$ has not been attempted (Gram–Schmidt over $\mathbb{Q}(q, T)$ is slow).

5. A dead end: the ι canonical basis

We tried the antilinear involution $\iota = \Psi \circ \beta$, where β is $(s, t) \mapsto (1/s, 1/t)$ on e -coefficients; it is an involution because $\Psi_{1/s, 1/t} = \Psi_{s, t}^{-1}$, and DS supplies unitriangularity. We hoped for a Lusztig-type canonical basis of $(\text{Sym}, \star)$ with positive structure constants. Lusztig’s lemma does produce a basis, since existence is formal. But the $\star$ structure constants in that basis have **mixed signs**, so there is no positivity and the idea is **dead**. We record it only for honesty.

6. Two questions for you

- Do you know whether Kirillov–Noumi (1999) states the $q \rightarrow \infty$ limit that we call Theorem A: on the $s = \infty$ edge of $\star$, $s^{-n(\mu)}e_{\mu_1} \star \cdots \star e_{\mu_\ell} \rightarrow P_{\mu'}(x; t)$ (Hall–Littlewood)? (Theorem B, the $t = 0$ edge, is already Di Francesco–Kedem 1505.01657 Cor. 5.18, as per my erratum.)
- What is your current review status of (N)? Unread, in progress, or checked? If checked, are there any steps you would downgrade?